

Spontaneous symmetry breaking

W. Ryssens



Summary of last time

- 1 Hamiltonian partitioning
- 2 Independent particle modelling
- 3 Ritz variational principle
- 4 Hartree-Fock equations of motion
- 5 Iterative strategy to tackle the problem
- 6 The Hartree-Fock energy and spectrum
- 7 The cost of doing business
- 8 Beyond Hartree-Fock

<https://www.nndc.bnl.gov/nudat3/>



Symmetries of the nuclear Hamiltonian

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 - linear, Hermitian ($U^\dagger = U$) $\rightarrow$ real eigenvalues
 - linear, Antihermitian ($U^\dagger = -U$) $\rightarrow$ imaginary eigenvalues

Labelling states

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Symmetry operation	Quantum number(s)	Operator U
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Gauge rotation	Particle numbers N & Z	$e^{-i[\alpha_n N_n + \alpha_p N_p]}$
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Time-reversal	None	$\check{T}$

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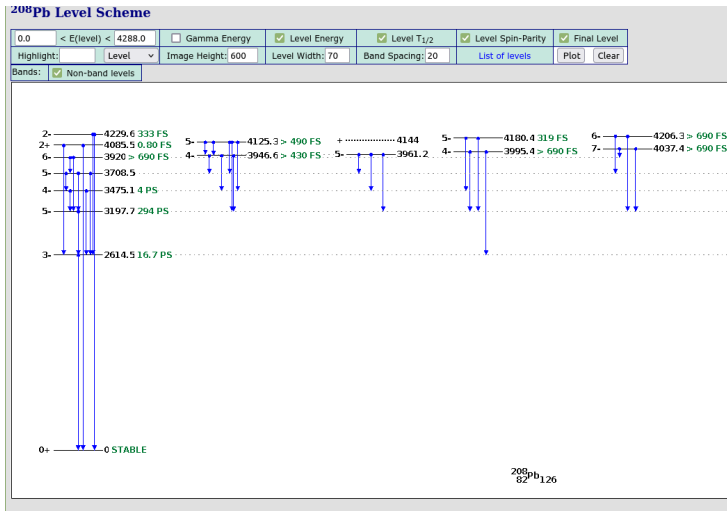
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Note: $\check{T}$ is the only antilinear operator in this list.

An example



Single-particle properties of symmetries

We will consider examples for two different sets of s.p. basis

- Plane wave states $|\vec{k}\sigma\rangle$
- Harmonic oscillator states $|n\ell jm\rangle$

Example 1: Parity

Basic observables

$$P\vec{r}P = -\vec{r}$$

$$P\vec{s}P = \vec{s}$$

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Example 2: Angular momentum

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Only mixes m quantum number!

$$e^{-i\vec{\alpha}\cdot\vec{J}}|nlj\,m\rangle = \sum_{m'} D_{mm'}^{j,*}(\Omega) |nlj\,m'\rangle$$

Wigner D-matrix Euler angles

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$$e^{-i\vec{\alpha}\cdot\vec{J}}a_{nljm}^\dagger e^{+i\vec{\alpha}\cdot\vec{J}} = \sum_{m'} D_{mm'}^{j,*}(\Omega) a_{nljm'}^\dagger.$$

Example 3: gauge rotation

Expressions for a single particle species; easily generalized to two species.

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$$e^{i\alpha N} a_{\vec{k}\sigma}^\dagger e^{-i\alpha N} = e^{i\alpha} a_{\vec{k}\sigma}^\dagger \qquad e^{i\alpha N} a_{nljm}^\dagger e^{-i\alpha N} = e^{i\alpha} a_{nljm}^\dagger.$$

Example 4: translation

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Symmetries and Slater determinants

The general transformation rule for creation and annihilation operators

Note: these are operator equations

$$a_{\alpha} \rightarrow a_{\bar{\alpha}} = U a_{\alpha} U^{\dagger} = \sum_{\beta} U_{\beta\alpha}^{*} a_{\beta} ,$$

$$a_{\alpha}^{\dagger} \rightarrow a_{\bar{\alpha}}^{\dagger} = U a_{\alpha}^{\dagger} U^{\dagger} = \sum_{\beta} U_{\beta\alpha} a_{\beta}^{\dagger} .$$

Transformed s.p. state $\uparrow$ $\uparrow$ $= \langle \beta | U | \alpha \rangle$

What about $\rho \rightarrow \bar{\rho}, E[\rho] \rightarrow E[\bar{\rho}], h[\rho] \rightarrow h[\bar{\rho}]$?

Symmetries of $E[\rho]$, ρ and h .

- For the density

$$\bar{\rho}_{ji} = \langle \Phi | U^\dagger a_i^\dagger U U^\dagger a_j U | \Phi \rangle = \sum_{\beta\gamma} U_{i\beta}^* U_{j\gamma} \langle \Phi | a_\beta^\dagger a_\gamma | \Phi \rangle = \sum_{\beta\gamma} U_{i\beta}^* U_{j\gamma} \rho_{\gamma\beta} .$$

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- The HF single-particle hamiltonian transforms

$$h[\bar{\rho}] = U h[\rho] U^\dagger.$$

Q for next time: prove this! Hint: use that $E[\rho] = E[\bar{\rho}]$ and vary both.

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This places strong constraints on the Slater determinants we can construct.
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Q: why do we have the freedom to do so?

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Symmetry	Shape of ρ^*	$\rho(\vec{r})$
Reflection	$\begin{pmatrix} \rho_+ & 0 \\ 0 & \rho_- \end{pmatrix}$	$\rho(-\vec{r}) = \rho(\vec{r})$
Translation	$\begin{pmatrix} \rho_{\vec{k}_1} & 0 & \dots \\ 0 & \rho_{\vec{k}_2} & \dots \\ \vdots & \vdots & \ddots \end{pmatrix}$	$\rho(\vec{r}) = \rho$

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Rotation	$\begin{pmatrix} \rho_{J=1/2} & 0 & \dots \\ 0 & \rho_{J=3/2} & \dots \\ \vdots & \vdots & \ddots \end{pmatrix}$	$\rho(\vec{r}) = \rho(r)$
	Matrix ↓	
	with $\rho_J = \begin{pmatrix} \rho_{J,M=-J} & 0 \\ 0 & \rho_{J,M=-J+1} \\ & & \ddots \end{pmatrix}$	
	and $\rho_{JM=-J} = \rho_{JM=-J+1} = \dots$	

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Extreme example

Demanding that

$$\vec{P}|\Psi\rangle = \vec{P}_{\text{cm}}$$

implies that

$$\rho(\vec{r}) = \rho$$

i.e. our model density is not localized!

Breaking symmetries

What if we allow our Slater determinants to mix representations?

Examples:

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- ✗ loss of quantum numbers

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- Rotation $\rightarrow |\Phi\rangle = \sum_{J=0,2,4,\dots} \sum_{|M|\leq J} c_{JM} |\theta_{JM}\rangle$.
- Parity $\rightarrow |\Phi\rangle = c_+ |\theta_+\rangle + c_- |\theta_-\rangle$
- Gauge rotation $\rightarrow |\Phi\rangle = \sum_{N=0,2,4,\dots} c_N |\theta_N\rangle$.

Pros

- ✓ larger variational space $\rightarrow$ (potentially) lower energy
- ✓ potential to describe specific excitation patterns

Cons

- ✗ loss of quantum numbers
...which is particularly problematic when dealing with selection rules

A state $|\Psi\rangle$ is **deformed** when it cannot be classified according to an **irreducible representation** of symmetry group of H .



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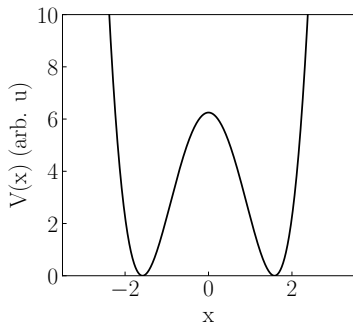
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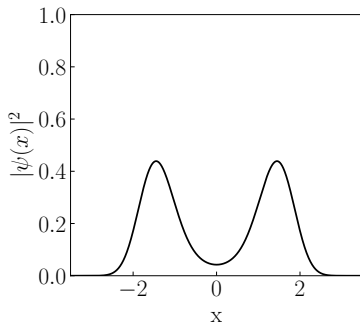
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Example: $|J = 2M = 2\rangle$ is not invariant under all rotations.

An **overly** simplified example



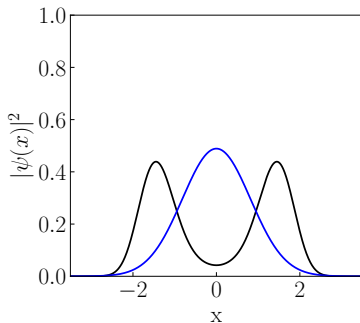
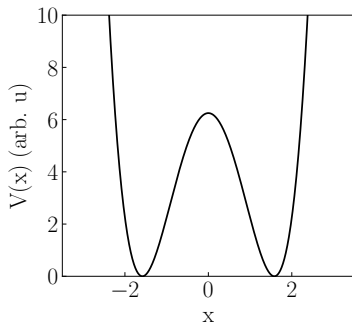
Exact



$\langle \hat{H} \rangle$
2.840

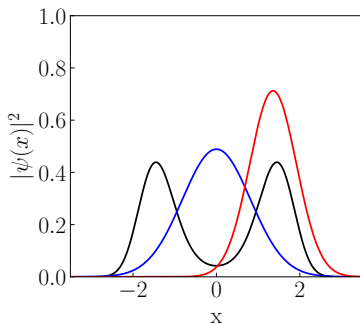
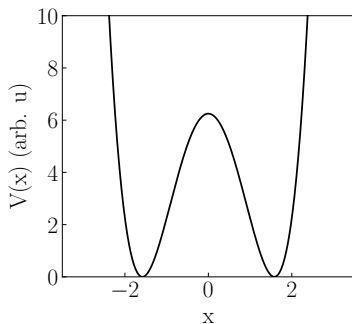
Dispersion Δx
2.02

An **overly** simplified example



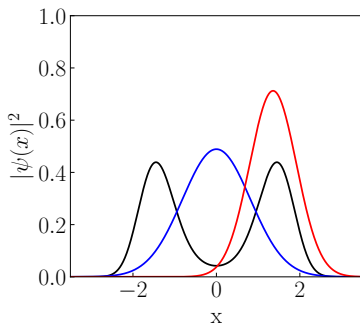
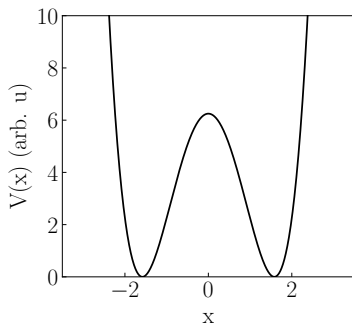
	$\langle \hat{H} \rangle$	Dispersion Δx
Exact	2.840	2.02
Symm. Gaussian	4.457	0.67

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Exact	2.840	2.02
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Warning: this one-particle example should not be driven too far!
It illustrates **symmetry breaking**, it **cannot** talk about collectivity.

Symmetries within HF calculation

Two possibilities in a practical calculation

- 1 at iteration (i) the symmetry is broken

$$\bar{\rho}^{(i)} \neq \rho^{(i)}$$

$$h[\bar{\rho}^{(i)}] = U h[\rho^{(i)}] U^\dagger.$$

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$\leftrightarrow$

A symmetry-broken solution can only be achieved with proper initialisation.

Intermezzo: characterizing the nuclear density

The **multipole moments** $Q_{\ell m}$ of the one-body density $\rho(\vec{r})$ are

$$\overbrace{Q_{\ell m}}^{\ell \geq 0, |m| \leq \ell} \equiv \int d^3r \rho(\vec{r}) r^\ell Y_{\ell m}(\theta, \phi),$$

↑ Spherical harmonics

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transformed to **dimensionless multipole moments** $\beta_{\ell m}$

$$\beta_{\ell m} \equiv \frac{4\pi}{3 \overset{R_0 = 1.2 \text{ fm}}{R_0}^\ell A^{1+\ell/3}} Q_{\ell m}.$$

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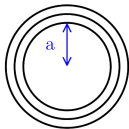
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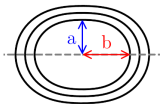
- These characterize the way a nuclear density is deformed
- For spherical $\rho(\vec{r})$, only $Q_{00} \sim N$ is non-zero.
- These are linked to electromagnetic moments.
- Conventions for $Y_{\ell m}$ and $\beta_{\ell m}$ are far from universal.

Visualisation

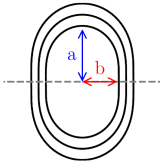
Spherical



Prolate



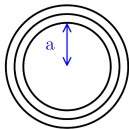
Oblate



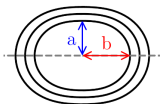
One DOF: β_{20}

Visualisation

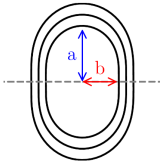
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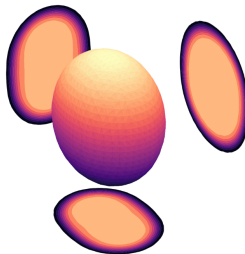


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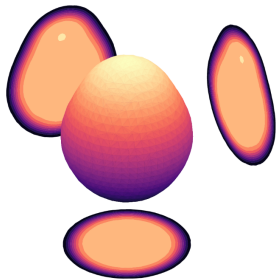


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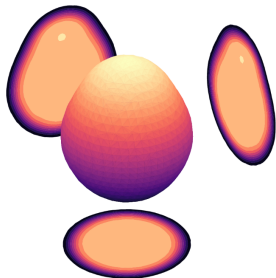
β_{20}, β_{22} or β_2, γ



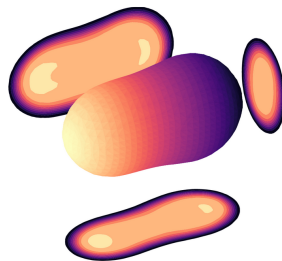
β_{20}, β_{30}



β_{20}, β_{30}



β_{20}, β_{22} **and** β_{30}



Symmetry breaking is associated with

- 1 non-zero order parameter(s) q , whose magnitude affects the energy
Ex: the dispersion of the associated quantum number $\Delta X \equiv \langle X^2 \rangle - \langle X \rangle^2$
- 2 a 'phase' that does not change the **mean-field** energy

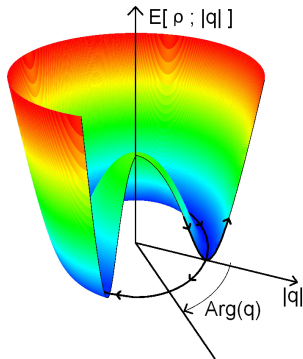
$$\text{because } E(\text{Arg}(q)) = \langle \phi | U^\dagger(\text{Arg}(q)) H U(\text{Arg}(q)) | \phi \rangle = \langle \phi | H | \phi \rangle .$$

Order parameters

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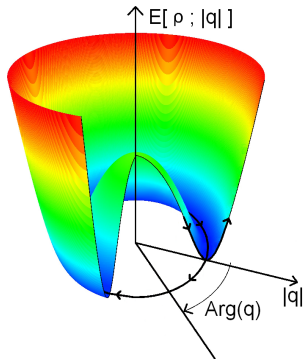


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Note: NOT comparable to the potential of our one particle example!

Order parameters

Symmetry	QNs	Order parameter(s) gained	Phases gained
Uniform	$N, J, M, \pi, \vec{P}_{\text{cm}}$	-	-
Spherical	N, J, M, π	$\Delta \vec{P}_{\text{cm}}^*$	$\vec{r}_{\text{CM}}$
Gauge	J, M, π	$\Delta N, \kappa_{\alpha\beta} \equiv \langle \Phi a_\beta a_\alpha \Phi \rangle$	Gauge angle ϕ
Axial	M, π	$\Delta J^2, \beta_{\ell 0}$ with ℓ even [†] , ...	1 Euler angle
Triaxial	π	$\Delta J_z, \beta_{\ell m}$ with ℓ, m even [†] , ...	2 Euler angles
Parity	-	$\Delta P, \beta_{\ell m}$ with m even [†] , ...	left $\leftrightarrow$ right

* $\rightarrow \Delta O = \langle O^2 \rangle - \langle O \rangle^2$.

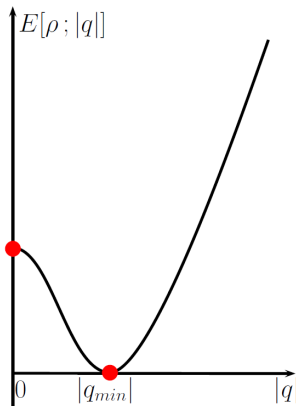
† $\rightarrow$ Selection rules on ℓ, m for g.s of even-even systems.

Correlation energy

Variation at $|q| \neq 0$

Variation at $|q| = 0$

$$E_{\text{corr}} = E_{\text{broken}}^{\text{HF}} - E_{\text{sym}}^{\text{HF}} \leq 0$$



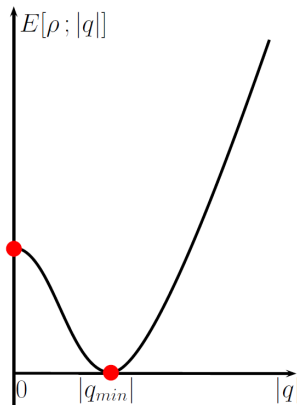
- Symmetry breaking
 - improves the energy w.r.t. to $E_{\text{exact}}^{\text{HF}}$
 - better starting point for more advanced methods
 - is 'strong' if E_{corr} is large
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 - is 'strong' if E_{corr} is large
 - is 'weak' if E_{corr} is small
- Minimize Routhian instead of energy

$$R = E - \lambda_q \langle \Phi | \hat{Q} | \Phi \rangle$$

to map $E(q)$!

Symmetry breaking allows us to model **collective motion**

- either by constructing a collective Hamiltonian

Rotational example: $H_{\text{coll}} = \sum_{\mu=x/y/z} \frac{J_{\mu}^2}{2\mathcal{M}}$

- ...or with a dedicated many-body technique
Example: Random phase approximation for vibrations

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Symmetry	Which?	Excitation patterns
Translation	All nuclei	Surface vibrations
Rotation	All except singly-magic	Rotational bands
Gauge	All except doubly-magic	Energy gap

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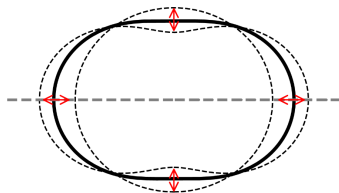
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Reminder

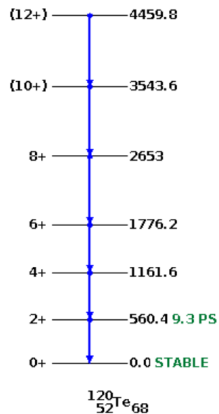
Real spectra mix all of these together and throw in particle-hole excitations!

Example 1a: translation

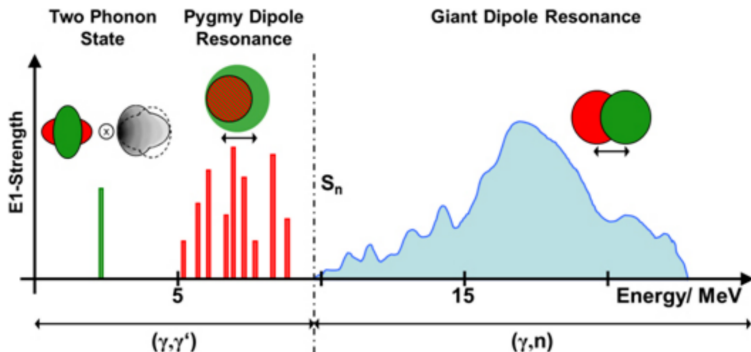
Theory



Experiment

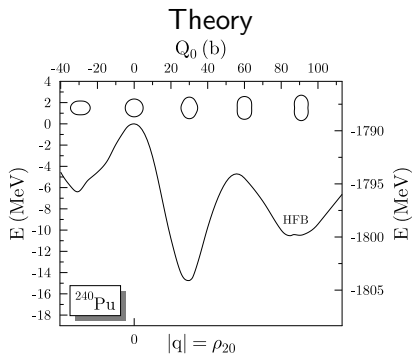


Example 1b: translation



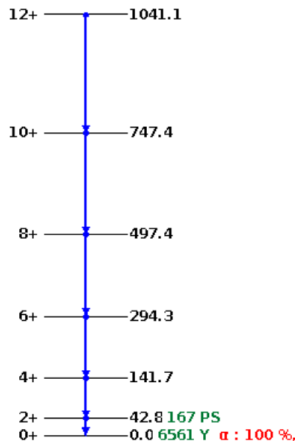
- Different giant resonances have been confirmed
 - Monopole, Dipole, Quadrupole, ...
 - Isovector, Isoscalar
 - With or without spin
- These dominate photon-absorption rates!

Example 2a: rotational symmetry



$\rho_{20} = \beta_{20}$ in our notation.

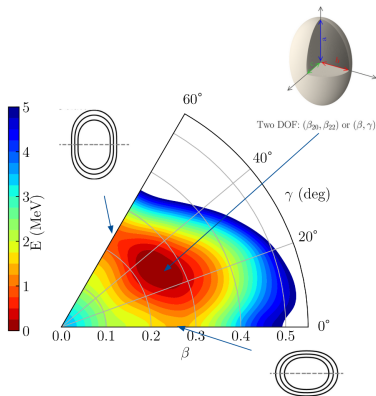
Experiment



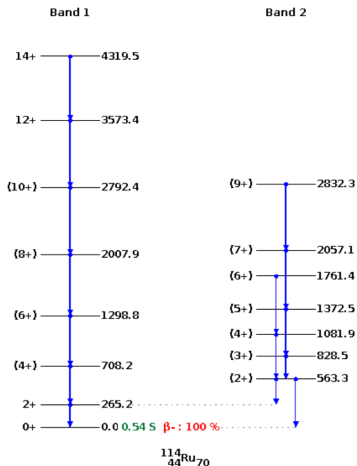
$^{240}_{94}\text{Pu}_{146}$

Example 2b: rotational symmetry

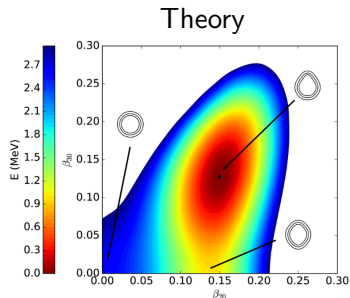
Theory



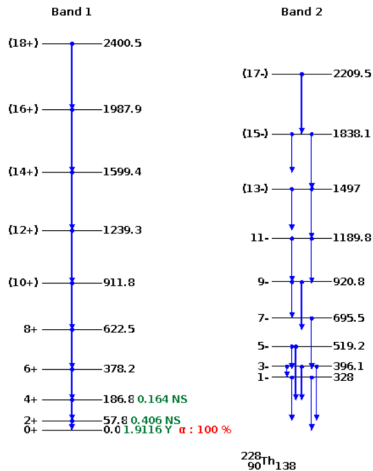
Experiment



Example 3: reflection symmetry



Experiment



Example 4: gauge rotation

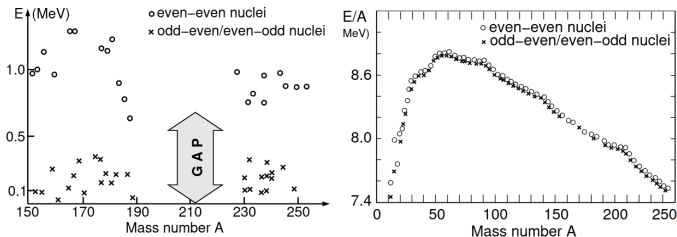
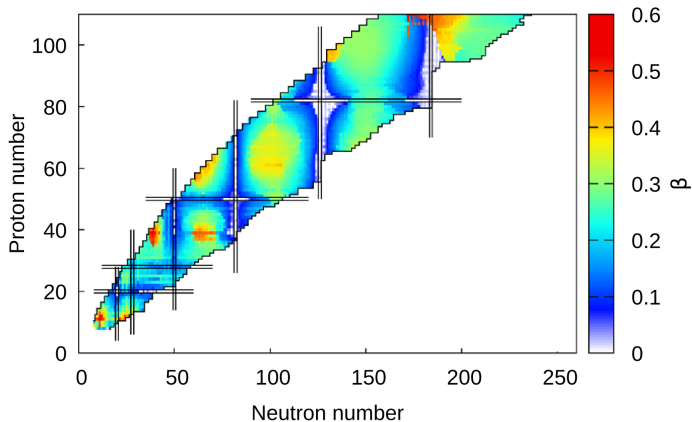


Figure of D. Page et al. arxiv:1302.6626 (2013).

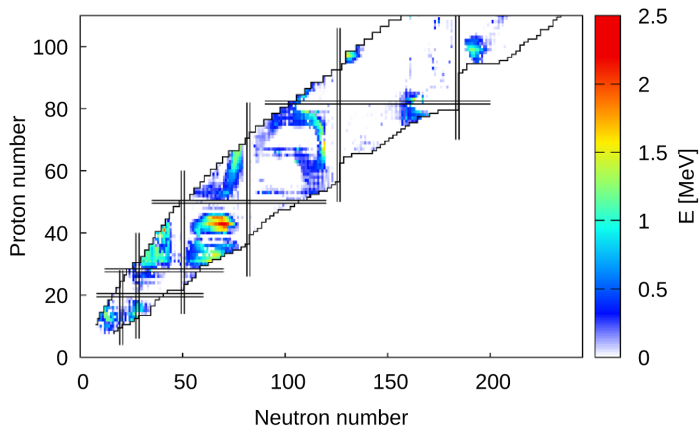
TBD next lecture!

Illustration: quadrupole deformation



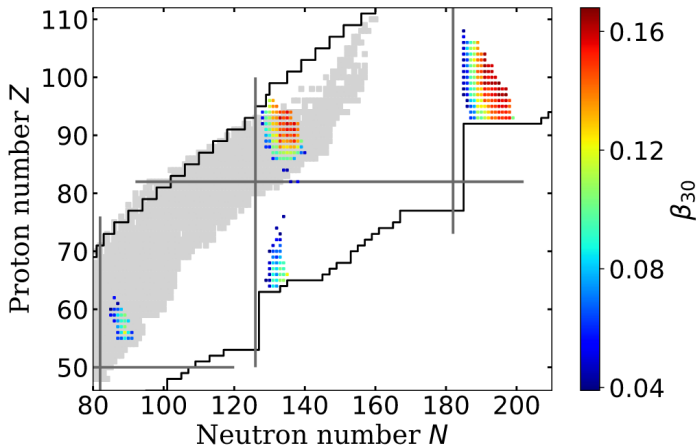
Total quadrupole deformation $\beta_2 = \sqrt{\beta_{20}^2 + 2\beta_{22}^2}$
according to the model of G. Scamps et al., EPJA 57 (2021).

Illustration: quadrupole deformation



Energy difference $E_{\text{axial}} - E_{\text{triaxial}}$
according to the model of G. Scamps et al., EPJA 57 (2021).

Illustration: octupole deformation



according to the model of G. Grams et al., EPJA 59 (2023).

Single-particle consequences

Broken	Dim $\rho(\vec{r})$	Quant. numb.	Degeneracy	
none	ρ	$\vec{k}, \sigma, q$	$\rho(E)$	

Remarks

- * ξ is a stand-in quantity to distinguish states related by time-reversal
- † rarely treated, remember isospin is not entirely a symmetry of H

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$\vec{p}, j^2, j_z, \pi, \mathcal{T}, T_z$	$\rho(r, \theta, \phi)$	n	1	†

Remarks

- * ξ is a stand-in quantity to distinguish states related by time-reversal
- † rarely treated, remember isospin is not entirely a symmetry of H

Single-particle consequences

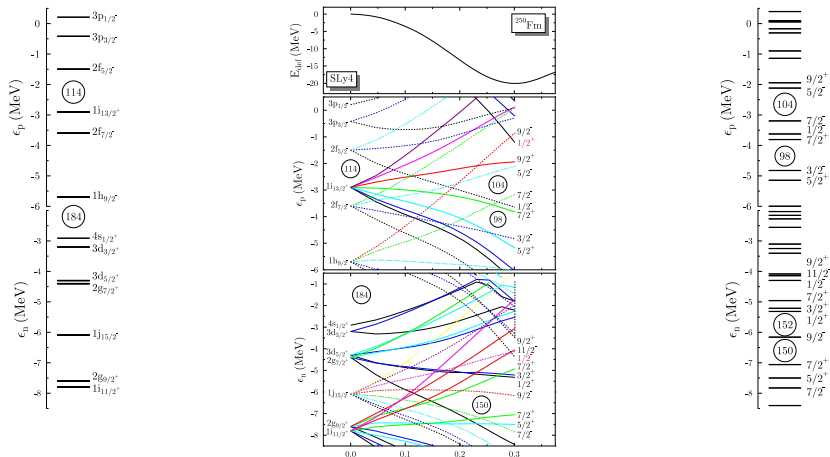
Broken	Dim $\rho(\vec{r})$	Quant. numb.	Degeneracy	
none	ρ	$\vec{k}, \sigma, q$	$\rho(E)$	
$\vec{p}$	$\rho(r)$	n, l, j, m, τ	$2j + 1$	
$\vec{p}, j^2$	$\rho(r, \theta)$	n, π, m, τ	2	
$\vec{p}, j^2, j_z$	$\rho(r, \theta, \phi)$	n, π, ζ, τ	2	*
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Remarks

- * ξ is a stand-in quantity to distinguish states related by time-reversal
- † rarely treated, remember isospin is not entirely a symmetry of H
- Single-particles determine computational complexity
- Less symmetries $\rightarrow$ **much** more computational time
- Less symmetries $\rightarrow$ less possibility to compare to experiment
- Less symmetries $\rightarrow$ less interpretation!

Deformation and Nilsson diagram

- Spectrum e_α as a function of deformation = Nilsson diagrams



[M. Bender, P.-H. Heenen, unpublished]₃₂

- β_{20} , is a compromise between $N = 150$ and $Z \approx 100$ deformed gaps
- Natural way to connect to empirical Nilsson model

How to do better: projection

Reminder: a deformed state mixes quantum numbers

$$\boxed{|\Psi\rangle} = \sum_{\alpha} \boxed{|\Psi_{\alpha}\rangle} .$$

Deformed Slater determinant Not a Slater determinant

Can we extract the component with desired quantum numbers?

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Can we extract the component with desired quantum numbers?

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$$P = \sum_{\alpha} |\Psi_{\alpha}\rangle \langle \Psi_{\alpha}|$$

We don't know the states $|\Psi_{\alpha}\rangle$ in advance, but **can we find projector operators such that**

$$P_{\alpha}^2 = P_{\alpha}?$$

Examples: parity (= a discrete symmetry)

Consider:

$$P_+ \equiv \frac{(1 + P)}{2},$$

$$P_- \equiv \frac{(1 - P)}{2}.$$

It is easy to check that

$$PP_{\pm} |\Psi\rangle = \pm P_{\pm} |\Psi\rangle.$$

(possibly) breaks P .

Not a Slater determinant!

Examples: gauge rotation (= a continuous symmetry)

Consider:

$$P_{N_0} = \frac{1}{2\pi} \int_0^{2\pi} d\phi_n \exp \left[i\phi_n \left(\overset{\text{Operator}}{\boxed{N}} - \underset{\text{Integer}}{\boxed{N_0}} \right) \right].$$

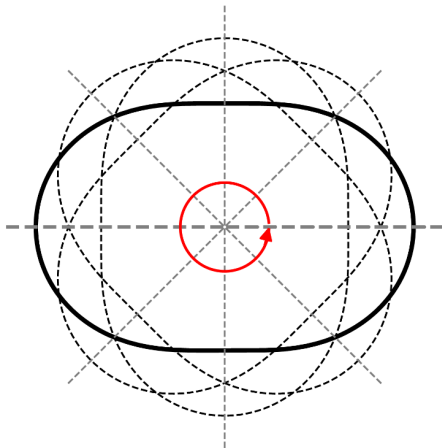
Check for next time:

$$NP_{N_0}|\Psi\rangle = N_0P_{N_0}|\Psi\rangle, \quad P_{N_0}^2 = P_{N_0},$$

Hints: (a) expand $|\Psi\rangle$ in states with good N and (b) do the left one first.

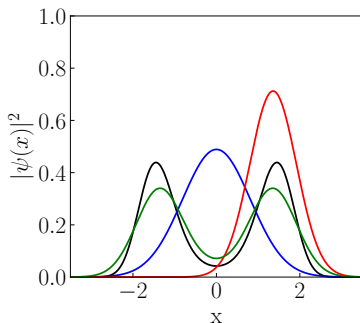
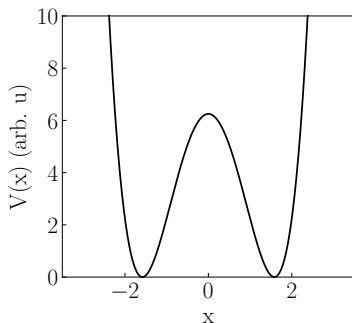
Examples: rotation

No formulas, they are hard* and numerically complicated.



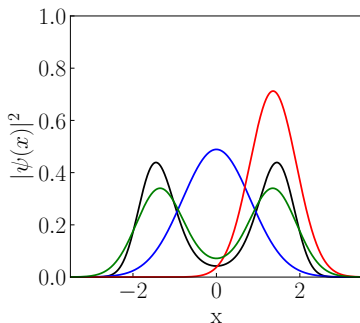
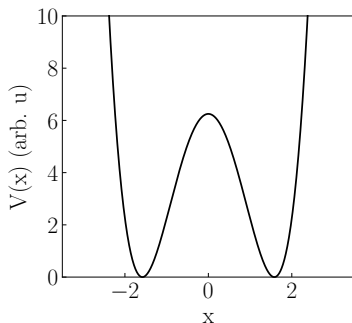
* $\rightarrow$ the Lie group behind rotations is non-Abelian, which causes trouble.

Back to our (overly) simple example



	$\langle \hat{H} \rangle$	Dispersion Δx
Exact	2.840	2.02
Symm. Gaussian	4.457	0.67
Asymm. Gaussian	2.994	2.16
Symmetry Restored	2.935	2.19

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Note how symmetry restoration **STILL** is not equal to the exact solution!

Discussion: are deformed states 'real'?

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This statement

- does not mean that its density resembles a rugby ball ...
- but does carry physical information, i.e. it is **useful!**

Exercise session 3 will be available by the end of the week.